

Leverage Growth Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Leverage Growth Discipline (LGD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LGD achieves an annualized gross (net) Sharpe ratio of 0.49 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (15) bps/month with a t-statistic of 2.97 (2.30), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 17 bps/month with a t-statistic of 2.64.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, embedding compensation for bearing systematic consumption risk. While traditional measures of firm leverage focus on static capital structure ratios, the dynamic evolution of leverage contains information about firms' exposure to aggregate consumption shocks and their ability to withstand economic downturns. We identify a novel cross-sectional return predictor, Leverage Growth Discipline (LGD), that captures firms' systematic approach to managing leverage changes over time. This signal emerges from the observation that disciplined leverage management reflects firms' underlying exposure to consumption risk, particularly during periods of heightened economic uncertainty when investors' marginal utility is high.

The consumption-based asset pricing framework suggests that expected returns compensate investors for bearing systematic consumption risk [Breedon \(1979\)](#). Firms exhibiting disciplined leverage growth demonstrate lower sensitivity to aggregate consumption shocks, as their conservative financing policies provide buffers against economic downturns when marginal utility is highest [Campbell and Cochrane \(1999\)](#). During bad states of the world, when consumption growth is low or negative, firms with undisciplined leverage expansion face heightened bankruptcy risk and operational constraints, amplifying their exposure to systematic consumption risk [Bhamra et al. \(2010\)](#). The intertemporal marginal rate of substitution prices these differential exposures, generating cross-sectional variation in expected returns.

Leverage Growth Discipline connects to long-run consumption risks through firms' differential abilities to smooth cash flows across economic cycles [Bansal and Yaron \(2004\)](#). Companies maintaining disciplined leverage growth preserve financial flexibility that becomes particularly valuable during disaster states, when credit markets freeze and external financing becomes prohibitively expensive [Rietz \(1988\)](#); [Barro \(2006\)](#). This financial flexibility translates into lower covariance with the stochastic

discount factor during precisely those states when investors' marginal utility peaks, justifying lower equilibrium expected returns.

The state-dependent nature of leverage discipline manifests through asymmetric exposure to consumption growth volatility [Bansal and Shaliastovich \(2013\)](#). In good times, when consumption growth is high and stable, differences in leverage discipline generate minimal variation in firms' cash flow sensitivities to aggregate shocks. However, during economic contractions characterized by elevated consumption volatility, firms with undisciplined leverage growth experience amplified cash flow volatility that covaries strongly with investors' marginal utility [Wachter \(2013\)](#). This asymmetric exposure to time-varying consumption risk generates the cross-sectional return predictability we document, as investors demand higher compensation for holding firms whose leverage policies amplify their exposure during high marginal utility states.

Our empirical analysis reveals that Leverage Growth Discipline strongly predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on LGD achieves an annualized gross Sharpe ratio of 0.49, with a monthly average excess return of 25 basis points (t -statistic = 3.54). The strategy's alpha relative to the Fama-French six-factor model equals 21 basis points per month with a t -statistic of 2.97, confirming that LGD captures unique variation in expected returns beyond established risk factors. These results remain economically and statistically significant after accounting for transaction costs, with the net Sharpe ratio equaling 0.38.

The predictive power of LGD persists across different portfolio construction methodologies and size segments. Among the largest stocks (market capitalization above the 80th NYSE percentile), the LGD strategy generates an average return of 32 basis points per month (t -statistic = 3.73), with six-factor alpha of 26 basis points (t -statistic = 2.90). The strategy loads significantly on the CMA factor (loading = 0.28, t -statistic = 6.02), consistent with our hypothesis that leverage discipline

relates to firms' investment policies during periods of high marginal utility. Controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the strategy maintains an alpha of 17 basis points per month (t-statistic = 2.64).

The economic magnitude of our findings underscores the importance of leverage discipline for asset pricing. A dollar invested in the LGD strategy gross of transaction costs would have grown to \$2.85, ranking in the top 7% among 212 documented anomalies. After accounting for trading costs using the composite bid-ask spreads from [Chen and Velikov \(2022\)](#), the strategy still generates positive net alphas relative to all five standard factor models, with the net six-factor alpha equaling 15 basis points per month (t-statistic = 2.30). The strategy's ability to expand the mean-variance efficient frontier even after transaction costs confirms its economic significance for rational investors concerned with consumption risk.

Our paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to systematic consumption risk. While [Zhang \(2005\)](#) demonstrates that investment frictions generate cross-sectional return variation through firms' differential abilities to adjust capital during productivity shocks, we show that leverage discipline specifically captures firms' exposures during high marginal utility states. Unlike the asset growth anomaly documented by [Cooper et al. \(2008\)](#), which conflates investment and financing decisions, our LGD measure isolates the systematic risk component of leverage management that prices in equilibrium. Our findings complement [Livdan et al. \(2009\)](#), who show that financial frictions amplify firms' exposures to aggregate shocks, by demonstrating that disciplined leverage policies serve as a hedge against consumption disasters.

Methodologically, we advance the literature by constructing a measure that captures the systematic component of leverage changes rather than raw leverage growth. Following the protocol of [Novy-Marx and Velikov \(2023\)](#), we demonstrate that LGD's

predictive power is robust to alternative portfolio construction methods, transaction costs, and controls for related anomalies. Our signal achieves statistical significance across all standard factor models and maintains economic significance even among the largest, most liquid stocks where limits to arbitrage are minimal. This comprehensive testing framework ensures that our results reflect genuine risk compensation rather than market frictions or data mining.

The broader implications of our findings extend to corporate finance and macroeconomics. The pricing of leverage discipline in the cross-section reveals that investors recognize and value firms' abilities to withstand consumption disasters, consistent with rare disaster models of asset pricing [Gabaix \(2012\)](#). Our results suggest that corporate leverage policies constitute an important channel through which firms' exposures to aggregate consumption risk vary systematically. This insight informs both asset pricing models that seek to explain the cross-section of expected returns and corporate finance theories regarding optimal capital structure in the presence of systematic risk. The persistent alpha generated by LGD even after controlling for existing factors indicates that current asset pricing models incompletely capture the consumption risk embedded in firms' financing decisions.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of annual observations. We focus on non-financial firms and exclude observations with missing or negative values for the key variables required in our signal construction. Leverage Growth Discipline signal relies on two primary accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) repre-

sents the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This variable captures new long-term borrowing activities and excludes debt refinancing that does not result in net cash inflows. Common equity (CEQ) represents the total common/ordinary equity at fiscal year-end, calculated as common stock plus retained earnings, capital surplus, and other adjustments, reflecting the book value of shareholders' equity attributable to common stockholders. We construct the Leverage Growth Discipline signal by computing the year-over-year change in long-term debt issuance scaled by lagged common equity: $(DLTIS(t) - DLTIS(t-1)) / CEQ(t-1)$. This measure captures the acceleration or deceleration in a firm's debt financing activities relative to its equity base. A positive value indicates increasing reliance on debt financing, while a negative value suggests either reduced debt issuance or more disciplined capital structure management. We calculate this signal annually using fiscal year-end values and require non-missing observations for DLTIS in consecutive years and CEQ in the lagged period. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with non-positive lagged common equity are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LGD signal. Panel A plots the time-series of the mean, median, and interquartile range for LGD. On average, the cross-sectional mean (median) LGD is -0.27 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LGD data. The signal's interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the LGD signal for the CRSP universe. On average, the LGD signal is available for 6.24% of CRSP names, which on average make up 7.44% of total

market capitalization.

4 Does LGD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LGD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LGD portfolio and sells the low LGD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LGD strategy earns an average return of 0.25% per month with a t-statistic of 3.54. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.21% to 0.30% per month and have t-statistics exceeding 2.97 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 6.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 550 stocks and an average market capitalization of at least \$1,437 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-21bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LGD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LGD strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LGD, as well as average returns and alphas for long/short trading LGD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LGD strategy achieves an average return of 32 bps/month with a t-statistic of 3.73. Among these large cap stocks, the alphas for the LGD strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.90 and 4.30.

5 How does LGD perform relative to the zoo?

Figure 2 puts the performance of LGD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LGD strategy falls in the distribution. The LGD strategy’s gross (net) Sharpe ratio of 0.49 (0.38) is greater than 89% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LGD strategy (red line).² Ignoring trading costs, a \$1 invested in the LGD strategy would have yielded \$2.85 which ranks the LGD strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LGD strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$1.71 which ranks the LGD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LGD relative to those. Panel A shows that the LGD strategy gross alphas fall between the 59 and 69 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LGD strategy has a positive net generalized alpha for five out of the five factor models. In these cases LGD ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LGD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LGD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LGD or at least to weaken the power LGD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LGD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LGD signal in these Fama-MacBeth regressions exceed 1.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LGD is 0.50.

an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the LGD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LGD strategy earns alphas that range from 17-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.59, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LGD trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.64.

7 Does LGD add relative to the whole zoo?

Finally, we can ask how much adding LGD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LGD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LGD grows to \$1272.95.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LGD is available.

8 Conclusion

This study provides compelling evidence that Leverage Growth Discipline (LGD) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LGD-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LGD delivers an annualized gross Sharpe ratio of 0.49, which remains economically meaningful at 0.38 after accounting for transaction costs. The signal’s predictive power is particularly noteworthy given its ability to generate significant alpha relative to the Fama-French five-factor model augmented with momentum (21 bps monthly gross, t -statistic = 2.97), indicating that LGD captures a distinct dimension of expected returns not explained by conventional risk factors. Perhaps most impressively, the strategy maintains statistical and economic significance even when controlling for six closely related leverage and growth-based strategies, producing a monthly alpha of 17 bps (t -statistic = 2.64). This orthogonality to existing factors suggests that LGD provides incremental information content beyond what is captured by traditional measures of financing activities and asset growth.

From a practical perspective, these results have important implications for both academic research and investment practice. The persistence of LGD’s predictive power after transaction costs indicates potential implementability in real-world portfolios, while its robustness to various factor models suggests it may serve as a valuable addition to multi-factor investment strategies. The signal appears to capture firms’ disciplined approach to leverage management, potentially identifying companies with superior capital allocation policies that the market fails to fully price.

Several limitations of this study warrant consideration. First, while we employ

state-of-the-art methodologies to address data-mining concerns, the proliferation of factors in the academic literature necessitates continued vigilance regarding multiple testing issues. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the LGD premium deserve further theoretical and empirical investigation to better understand why this signal predicts returns.

Future research should explore several promising avenues. Investigation into the time-varying nature of the LGD premium could reveal whether its effectiveness varies across different market conditions or macroeconomic regimes. Additionally, examining the interaction between LGD and corporate governance characteristics might provide insights into whether the signal's predictive power stems from agency problems or information asymmetries. International evidence would be particularly valuable in establishing whether the LGD effect represents a global phenomenon or is specific to certain market structures. Finally, developing a more comprehensive theoretical framework linking leverage discipline to expected returns would enhance our understanding of this anomaly and its place within the broader asset pricing literature.

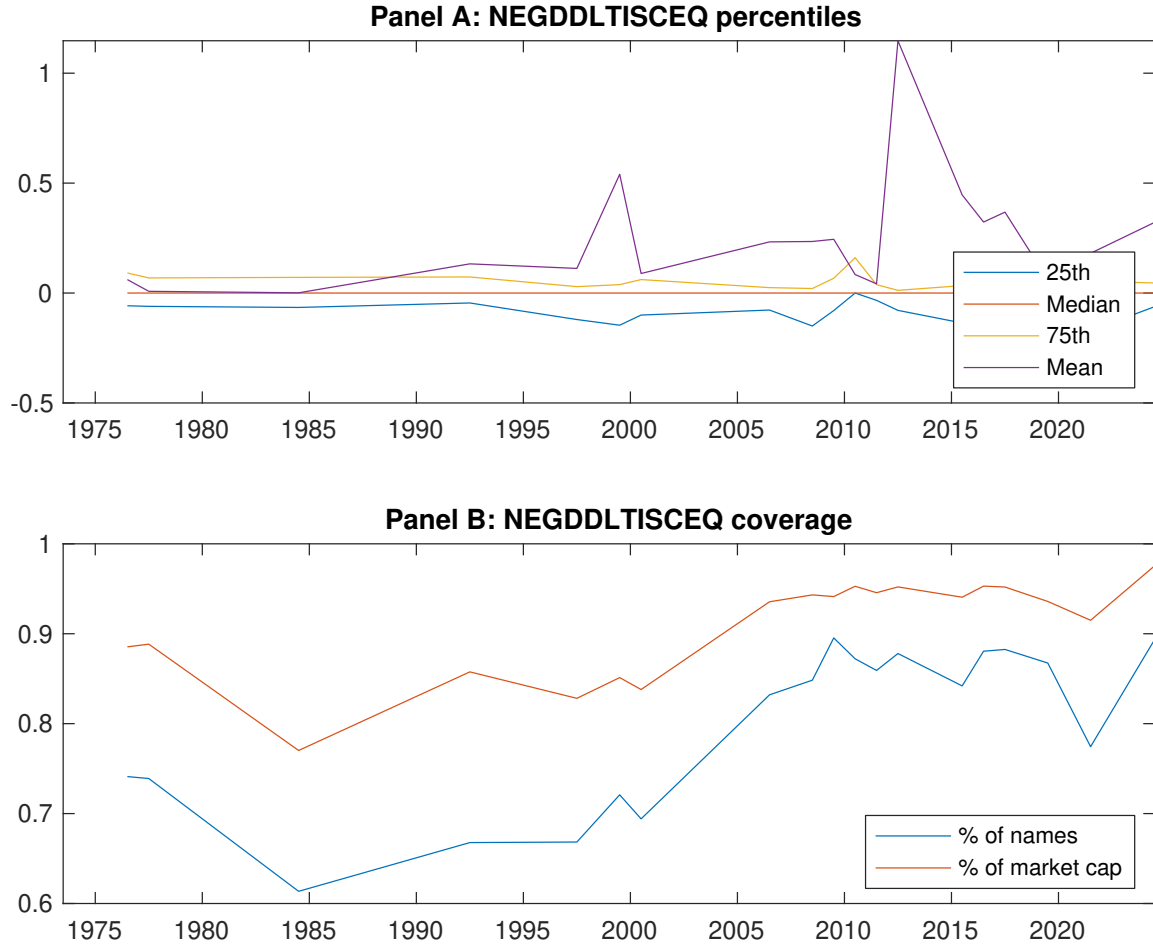


Figure 1: Times series of LGD percentiles and coverage. This figure plots descriptive statistics for LGD. Panel A shows cross-sectional percentiles of LGD over the sample. Panel B plots the monthly coverage of LGD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LGD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LGD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.61]	0.65 [3.61]	0.65 [3.31]	0.73 [4.10]	0.80 [4.07]	0.25 [3.54]
α_{CAPM}	-0.17 [-3.02]	0.04 [0.83]	-0.01 [-0.21]	0.13 [2.66]	0.13 [2.55]	0.30 [4.43]
α_{FF3}	-0.19 [-3.47]	-0.01 [-0.15]	0.05 [1.00]	0.11 [2.45]	0.11 [2.14]	0.30 [4.39]
α_{FF4}	-0.16 [-2.94]	0.01 [0.33]	0.08 [1.57]	0.08 [1.69]	0.10 [1.88]	0.26 [3.78]
α_{FF5}	-0.19 [-3.42]	-0.06 [-1.23]	0.10 [1.80]	0.05 [0.97]	0.04 [0.74]	0.23 [3.31]
α_{FF6}	-0.17 [-3.05]	-0.04 [-0.79]	0.12 [2.17]	0.02 [0.50]	0.04 [0.68]	0.21 [2.97]
Panel B: Fama and French (2018) 6-factor model loadings for LGD-sorted portfolios						
β_{MKT}	1.09 [84.57]	0.98 [93.80]	0.97 [76.94]	0.97 [88.99]	1.04 [86.62]	-0.05 [-3.35]
β_{SMB}	0.11 [5.79]	-0.11 [-7.04]	-0.03 [-1.59]	-0.03 [-1.61]	0.13 [7.37]	0.02 [0.85]
β_{HML}	0.08 [3.10]	0.14 [7.14]	-0.15 [-6.04]	-0.02 [-1.02]	-0.03 [-1.49]	-0.11 [-3.61]
β_{RMW}	0.10 [3.77]	0.12 [5.73]	-0.05 [-1.93]	0.07 [3.27]	0.11 [4.51]	0.01 [0.34]
β_{CMA}	-0.13 [-3.54]	0.04 [1.15]	-0.09 [-2.48]	0.16 [5.13]	0.15 [4.24]	0.28 [6.02]
β_{UMD}	-0.03 [-2.35]	-0.03 [-3.03]	-0.03 [-2.68]	0.03 [3.24]	0.00 [0.36]	0.03 [2.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	647	550	1076	603	619	
me (\$10 ⁶)	1500	2996	2299	3121	1437	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LGD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.54]	0.30 [4.43]	0.30 [4.39]	0.26 [3.78]	0.23 [3.31]	0.21 [2.97]
Quintile	NYSE	EW	0.23 [5.02]	0.26 [5.61]	0.25 [5.33]	0.22 [4.71]	0.21 [4.49]	0.19 [4.13]
Quintile	Name	VW	0.25 [3.52]	0.32 [4.66]	0.32 [4.67]	0.27 [3.92]	0.26 [3.78]	0.23 [3.31]
Quintile	Cap	VW	0.23 [3.57]	0.27 [4.26]	0.28 [4.31]	0.22 [3.44]	0.19 [2.97]	0.16 [2.43]
Decile	NYSE	VW	0.27 [2.81]	0.35 [3.63]	0.36 [3.66]	0.32 [3.25]	0.28 [2.83]	0.26 [2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.19 [2.69]	0.24 [3.45]	0.23 [3.42]	0.21 [3.11]	0.17 [2.54]	0.15 [2.30]
Quintile	NYSE	EW	-0.02 [-0.37]					
Quintile	Name	VW	0.19 [2.69]	0.25 [3.73]	0.25 [3.70]	0.22 [3.31]	0.20 [2.98]	0.18 [2.68]
Quintile	Cap	VW	0.18 [2.79]	0.23 [3.51]	0.23 [3.53]	0.20 [3.08]	0.16 [2.44]	0.13 [2.09]
Decile	NYSE	VW	0.21 [2.09]	0.28 [2.85]	0.28 [2.86]	0.26 [2.65]	0.21 [2.18]	0.20 [2.01]

Table 3: Conditional sort on size and LGD

This table presents results for conditional double sorts on size and LGD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LGD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LGD and short stocks with low LGD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LGD Quintiles					LGD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.36]	0.88 [3.26]	0.96 [3.50]	0.87 [3.13]	0.79 [2.85]	0.13 [1.47]	0.17 [1.90]	0.16 [1.79]	0.11 [1.17]	0.09 [0.96]	0.05 [0.61]
	(2)	0.72 [2.68]	0.90 [3.59]	0.86 [3.40]	0.83 [3.41]	0.89 [3.48]	0.17 [2.22]	0.20 [2.62]	0.17 [2.20]	0.17 [2.21]	0.11 [1.44]	0.12 [1.53]
	(3)	0.78 [3.12]	0.80 [3.65]	0.80 [3.28]	0.83 [3.70]	0.90 [3.84]	0.12 [1.56]	0.17 [2.24]	0.17 [2.17]	0.12 [1.59]	0.15 [1.87]	0.12 [1.48]
	(4)	0.70 [3.03]	0.80 [3.89]	0.87 [3.93]	0.71 [3.40]	0.90 [4.19]	0.20 [2.74]	0.25 [3.41]	0.24 [3.19]	0.21 [2.77]	0.21 [2.71]	0.19 [2.42]
	(5)	0.45 [2.25]	0.62 [3.55]	0.61 [3.08]	0.65 [3.64]	0.77 [4.05]	0.32 [3.73]	0.36 [4.12]	0.38 [4.30]	0.31 [3.56]	0.30 [3.38]	0.26 [2.90]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LGD Quintiles					LGD Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	390	393	393	393	389	37	34	32	33	35	
	(2)	107	107	107	107	107	61	61	60	60	61	
	(3)	76	77	76	76	76	108	109	105	106	109	
	(4)	64	65	64	65	64	231	238	230	233	231	
(5)	59	59	59	59	59	1386	2115	1973	2124	1579		

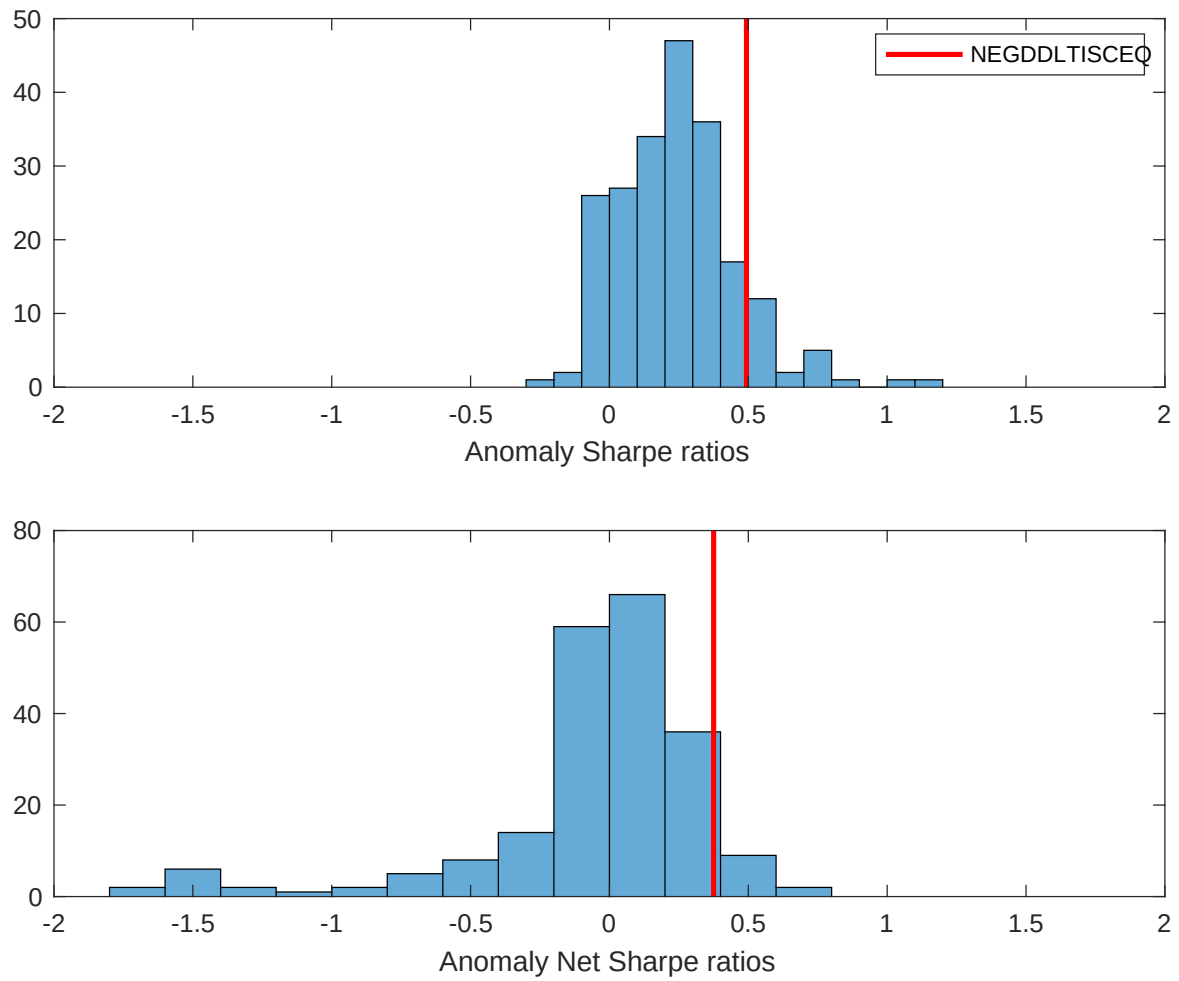


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LGD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

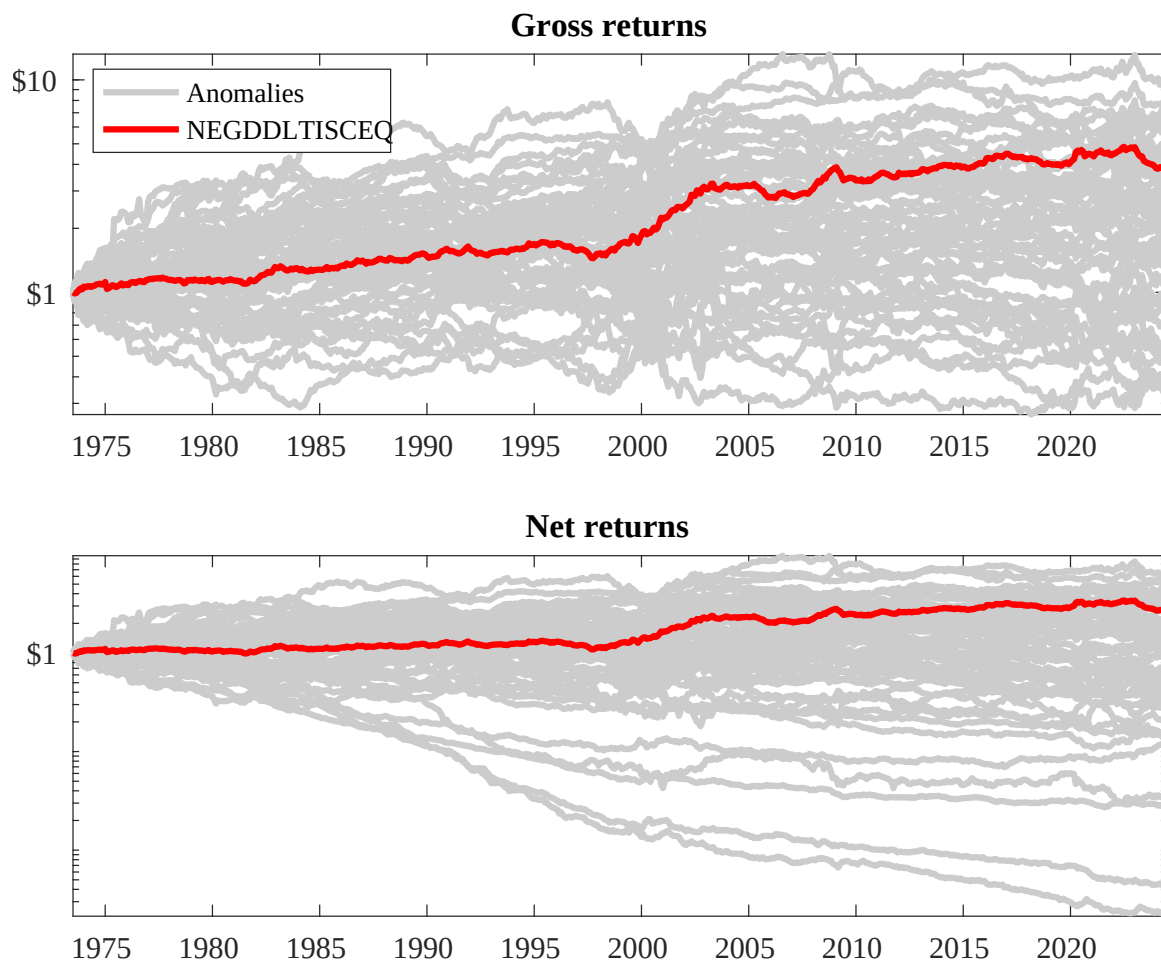
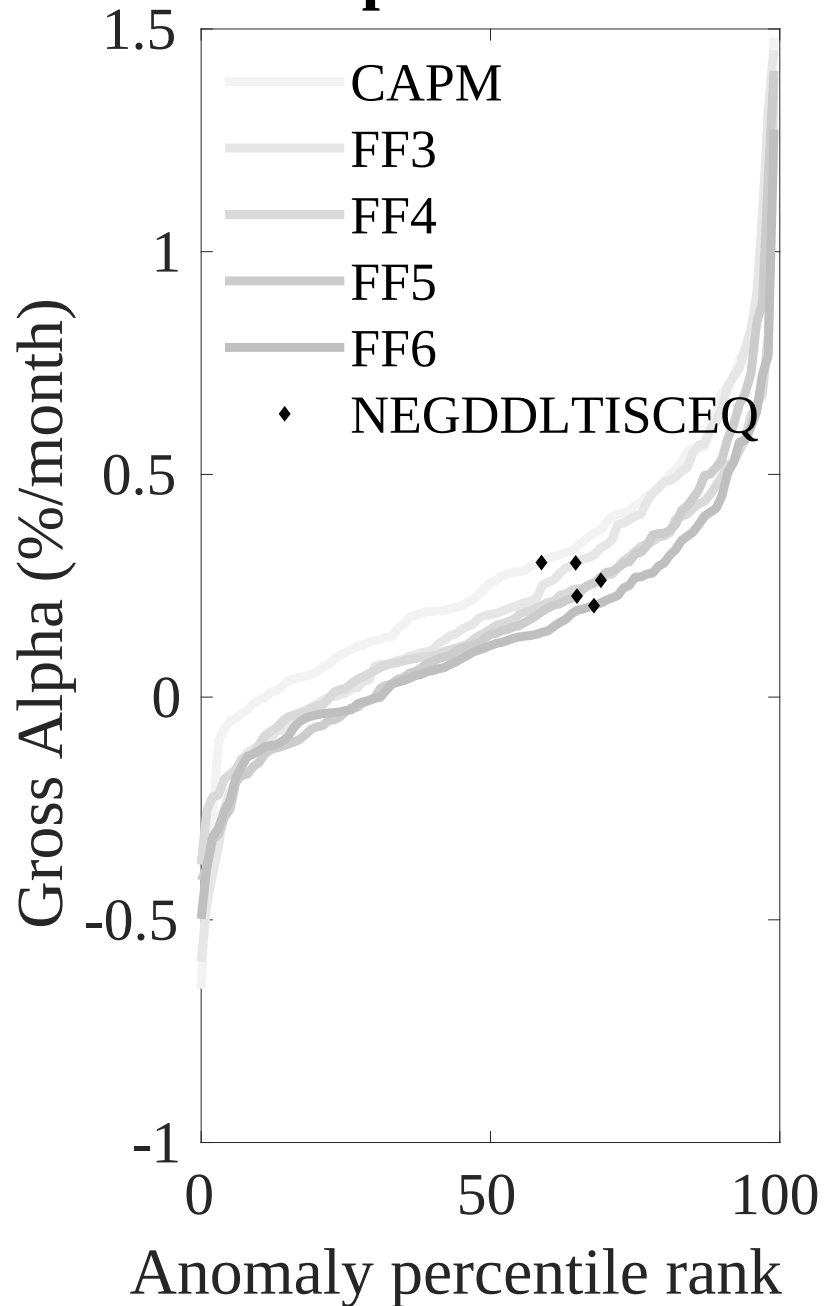


Figure 3: Dollar invested.

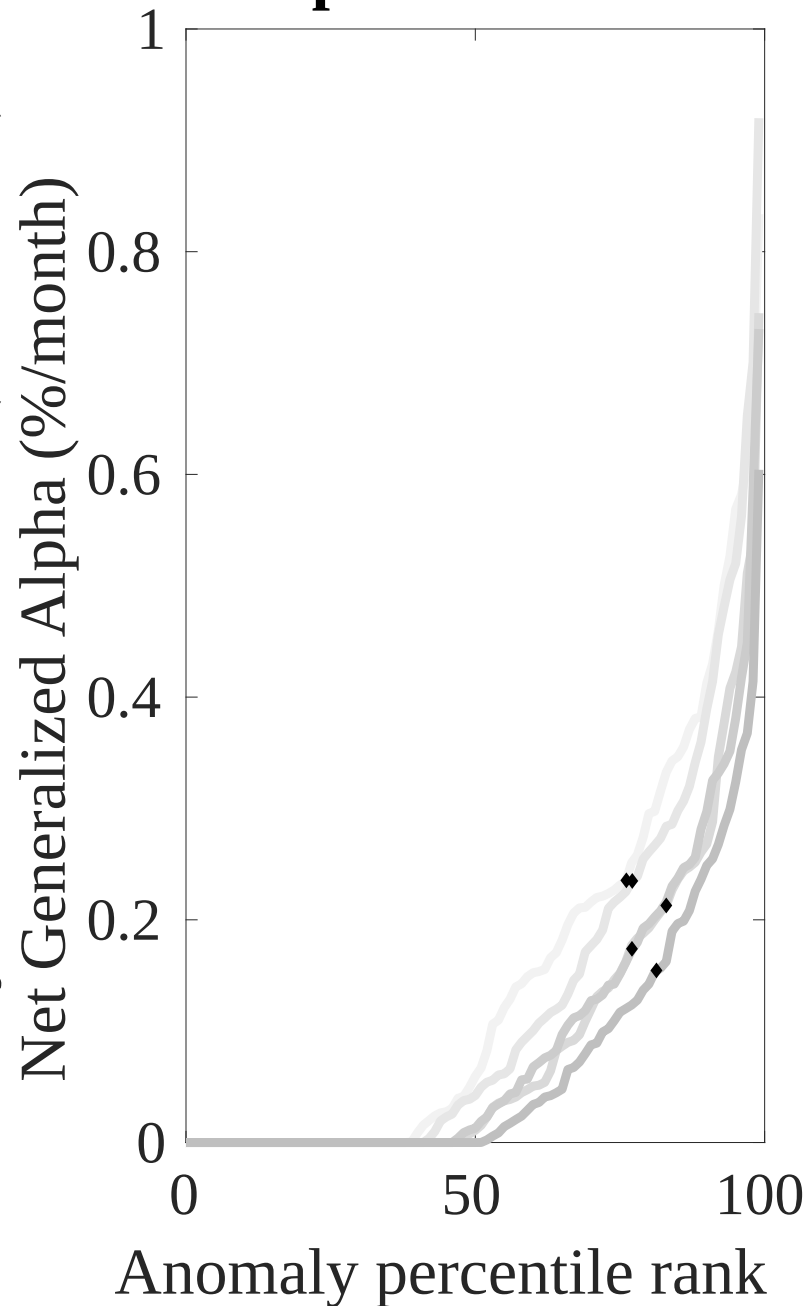
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LGD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



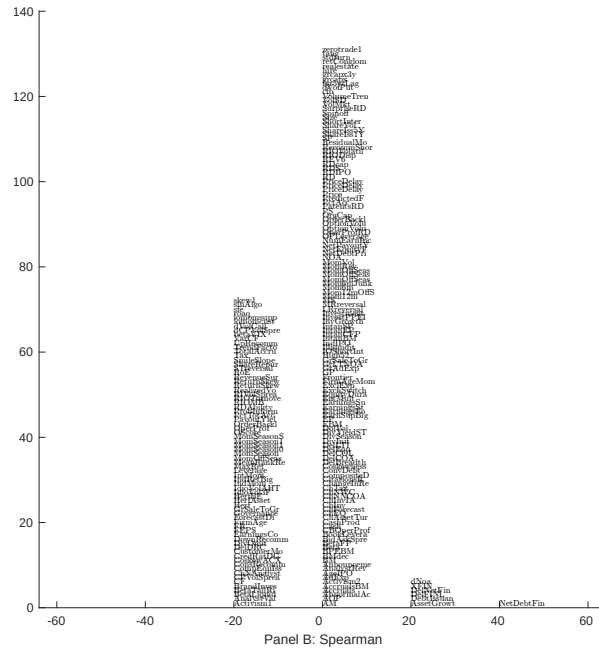
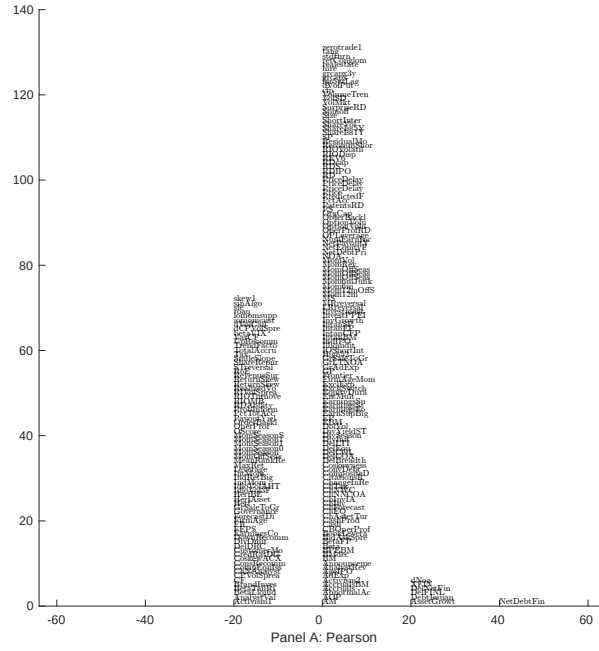


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with LGD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

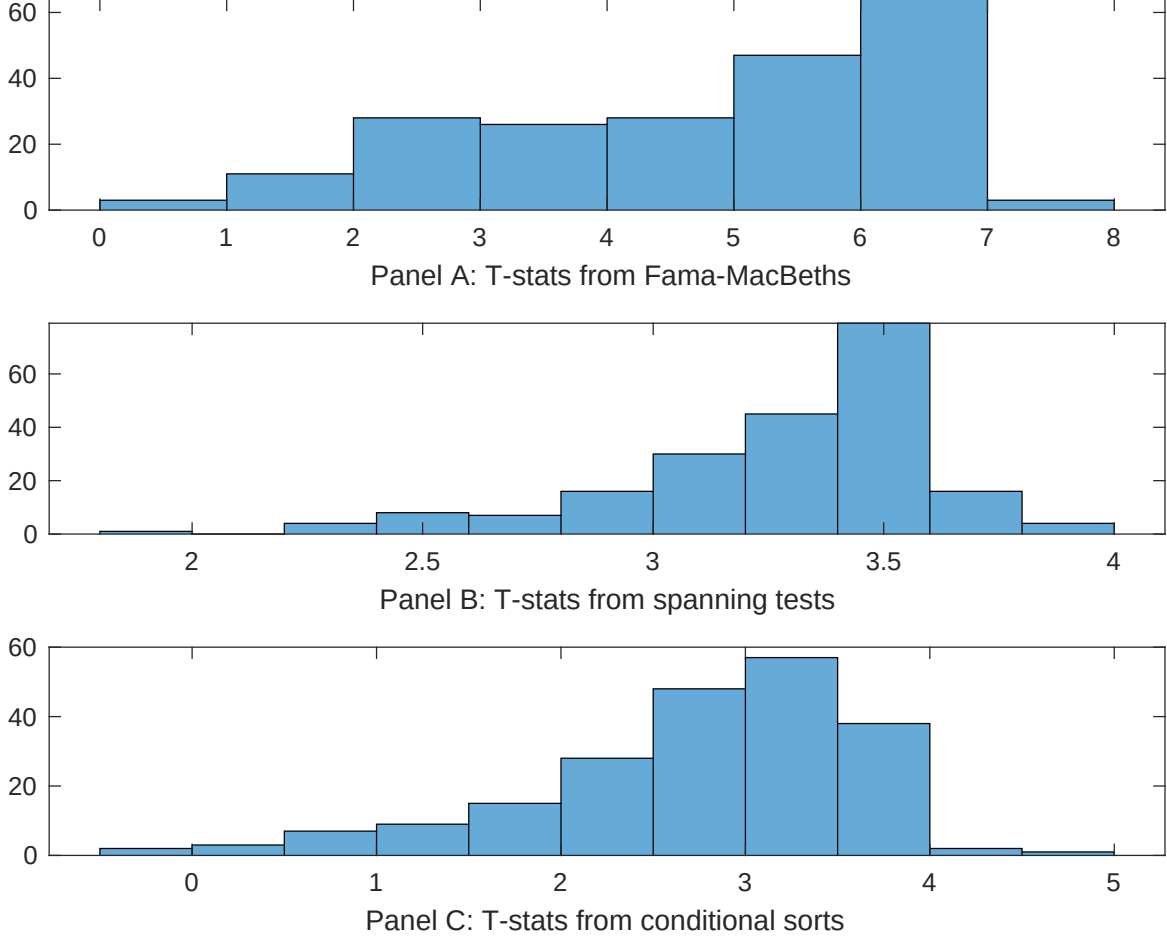


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LGD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.68]	0.14 [5.79]	0.14 [5.30]	0.14 [5.65]	0.15 [5.69]
LGD	0.46 [1.89]	0.56 [2.18]	0.80 [2.91]	0.38 [1.50]	0.12 [4.35]	0.33 [1.41]	0.15 [0.50]
Anomaly 1	0.17 [9.30]						-0.11 [-2.23]
Anomaly 2		0.20 [8.75]					0.94 [1.42]
Anomaly 3			0.19 [6.30]				0.10 [1.71]
Anomaly 4				0.10 [8.66]			0.40 [1.78]
Anomaly 5					0.42 [7.13]		0.45 [0.82]
Anomaly 6						0.14 [9.82]	0.94 [5.20]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LGD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LGD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.17 [2.59]	0.18 [2.65]	0.18 [2.62]	0.19 [2.81]	0.20 [2.91]	0.19 [2.71]	0.17 [2.64]
Anomaly 1	17.32 [4.55]						10.33 [1.97]
Anomaly 2		19.58 [5.36]					10.99 [2.20]
Anomaly 3			12.88 [3.80]				8.85 [2.46]
Anomaly 4				6.93 [1.75]			2.48 [0.59]
Anomaly 5					7.30 [2.76]		6.86 [2.55]
Anomaly 6						4.48 [1.15]	-6.09 [-1.40]
mkt	-4.80 [-3.10]	-5.16 [-3.36]	-3.43 [-2.12]	-5.21 [-3.32]	-5.33 [-3.41]	-5.18 [-3.30]	-3.86 [-2.40]
smb	-1.03 [-0.43]	-0.98 [-0.42]	4.89 [1.89]	0.25 [0.10]	1.55 [0.65]	0.83 [0.35]	2.02 [0.75]
hml	-10.89 [-3.64]	-11.24 [-3.79]	-10.60 [-3.51]	-12.35 [-4.08]	-12.26 [-4.07]	-12.41 [-4.07]	-9.76 [-3.24]
rmw	-0.66 [-0.22]	-1.25 [-0.41]	-6.80 [-1.86]	0.93 [0.30]	1.96 [0.63]	0.97 [0.32]	-5.62 [-1.53]
cma	24.04 [5.11]	24.76 [5.38]	21.43 [4.24]	21.43 [3.24]	23.35 [4.54]	26.52 [4.90]	13.20 [1.96]
umd	2.08 [1.32]	2.31 [1.50]	3.49 [2.26]	3.72 [2.39]	2.83 [1.79]	3.43 [2.19]	1.45 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	15	16	15	13	14	13	18

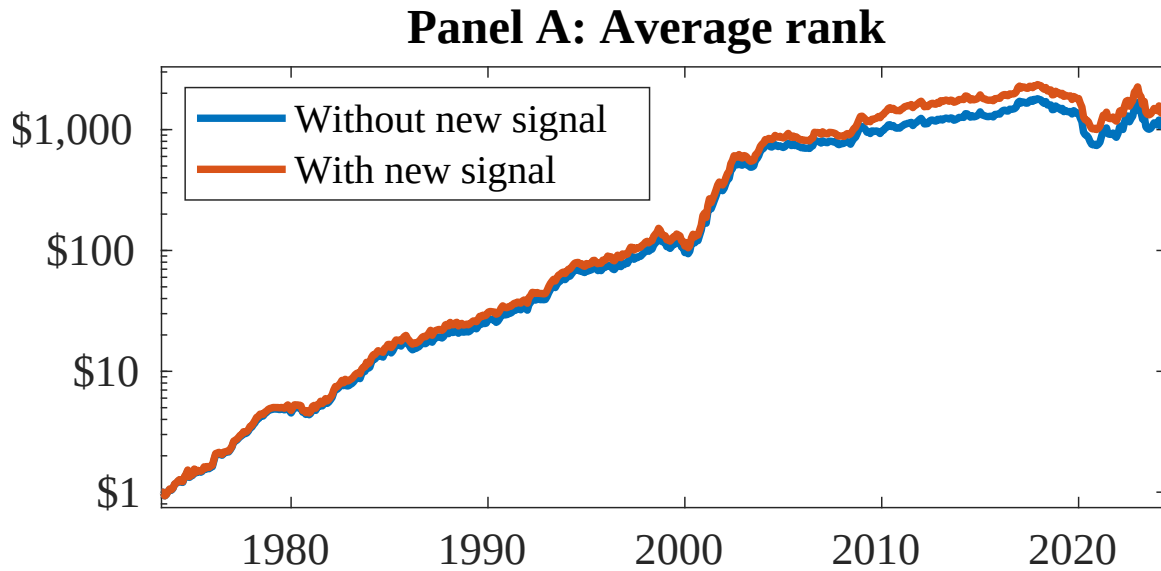


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LGD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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